SAC Revision — Unit 4 Outcome 3

VCE Algorithmics (HESS) · Computer science: past and present

SAC: Friday 11 September · 100 minutes · 50 marks · structured questions.
Solutions to practice questions discussed in class Wednesday.

Part 1 — The checklist

Tick only when you can do it closed-book. These are the outcome's dot points — every SAC question maps to one.

History and computability

☐ Explain the historical context for the emergence of computer science (Hilbert → Entscheidungsproblem → Turing/Church 1936).
☐ Describe the general structure of a Turing machine (tape, head, states, transition table).
☐ Trace a small Turing machine by hand from a rule table.
☐ State the Church–Turing thesis and explain why it is a thesis, not a theorem.
☐ Reproduce the Halting Problem undecidability proof (assume H → build D → run $D(D)$ → contradiction).
☐ Apply undecidability to real claims (loop detectors, program equivalence, malware detection).

Conceptions of AI

☐ Describe the Turing Test and the question it replaces.
☐ Distinguish strong AI from weak AI, and identify which the Chinese Room targets.
☐ Describe the Chinese Room accurately (setup, syntax-vs-semantics point, conclusion).
☐ Mount a case for or against it, engaging one standard reply (systems / robot) with a counter.

Machine learning

☐ Explain how a data-driven algorithm learns (parameters, error, iteration).
☐ State the SVM training objective (maximise the margin; support vectors).
☐ Explain and demonstrate the higher-dimensions trick for non-linearly-separable data.
☐ Describe the structure of a multi-layer perceptron (neurons, weights, biases, activations, layers).
☐ Evaluate a small MLP by forward propagation, showing all sums.
☐ Define overfitting and underfitting via training-vs-new-data error, with one fix each.
☐ Discuss an AI ethics scenario: stakeholder → harm → mechanism → mitigation.

Part 2 — Practice questions (SAC format)

P1. (3 marks) Explain why Hilbert's program needed a formal definition of "algorithm" before the Entscheidungsproblem could be answered in the negative.

P2. (4 marks) A Turing machine in state q_0, head on the leftmost symbol of 110, has rules: $(q_0, 1) \to (1, R, q_0)$; $(q_0, 0) \to (1, L, q_1)$; $(q_1, 1) \to (0, L, q_1)$; $(q_1, \square) \to (\square, S, H)$. Trace it and give the final tape.

P3. (5 marks) Reproduce the Halting Problem proof. Structure: assumption, construction, self-application, case analysis, conclusion.

P4. (2 marks) Your IDE flags: "possible infinite loop — cannot verify." Why is "cannot verify" the honest wording? Refer to decidability.

P5. (7 marks) "The Chinese Room shows that no computer running a program can genuinely understand language." Describe the argument this claim rests on, then argue for or against it, engaging the systems reply.

P6. (3 marks) Class A: $(1, 1), (3, 3)$. Class B: $(1, 3), (3, 1)$. Show no separating line exists and construct a feature that fixes it.

P7. (4 marks) Evaluate by forward propagation, step activations, showing working: inputs $x_1 = 0, x_2 = 1$; h_1: weights $(0.3, 0.5)$, bias -0.6; h_2: weights $(0.8, -0.2)$, bias 0.1; output: weights $(1.0, 0.5)$ from (h_1, h_2), bias -0.4.

P8. (2 marks) Training error 1%, test error 40%: name the problem and give two distinct remedies.

P9. (4 marks) A bank's loan model, trained on 20 years of approvals, is found to decline applicants from certain suburbs at twice the rate of otherwise-identical applicants. Write the four-move ethics paragraph (stakeholder, harm, mechanism, mitigation).

Part 3 — Quick answers for self-checking

P2: head runs right over the two 1s, hits 0 → writes 1, moves L into q_1; q_1 zeroes the 1s leftward, halts at blank. Final tape: 001.

P6: diagonals share midpoint $(2, 2)$, so no line; $z = |x - y|$ gives A: 0,0 and B: 2,2 — separable by $z = 1$.

P7: h_1: $0.5 - 0.6 = -0.1 \leq 0 \Rightarrow 0$; h_2: $-0.2 + 0.1 = -0.1 \leq 0 \Rightarrow 0$; out: $0 + 0 - 0.4 \leq 0 \Rightarrow$ **0**.

P8: overfitting; more data / simpler model / held-out evaluation.

P1, P3, P4, P5, P9: use the checklist structures; model answers discussed Wednesday.